BT4222 Group 13 — Dataset & Source-Code Listing

GitHub repository: https://github.com/steffanylars/nus-web-mining-business-insights (branch: steff-coldstart-diversity)

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1. Datasets

JDSearch (Liu et al., SIGIR 2023) — 173,831 users, 12.8M products, 26.7M interactions. Source and download (JD Cloud): https://github.com/rucliujn/JDsearch · paper https://doi.org/10.1145/3539618.3591900. Raw files are not redistributed (licence and size): user_behavior_data.txt (957 MB) and product_meta_data.txt (4.8 GB). Set JD_USER_TXT / JD_PRODUCT_TXT or place them in Final Project/data/.

Cleaned / intermediary artifacts. Regenerated deterministically by stages 2–3 (seed 42; split hash recorded in artifacts/corrected/dataset_manifest.json): candidates.parquet, history.parquet, sessions.parquet, meta_subset.parquet, features.parquet. These are git-ignored for size (~360 MB); a copy is shared at the Drive link below. Lightweight measured outputs (predictions, metrics, model files) are committed under artifacts/corrected and artifacts/step7_draft_artifacts.

Intermediate-artifact share link: <<INSERT Google Drive / OneDrive link before submission>>

2. Repository structure

Path	Contents
README.md	Repository overview and pointers.
Interim Report/JDSearch_Data_Anlaysis.ipynb	Interim EDA notebook (data acquisition, descriptive statistics, feature dictionary).
Final Project/README.md	Task, protocol, reproduction commands (run_pipeline.py, env vars), stage list.
Final Project/ETHAN_CHANGES.md	Audit change log: what was corrected in the pipeline and why; corrected vs draft scores.
Final Project/STEP7_COLDSTART_DIVERSITY.md	Cold-start routing and category diversification: implementation, measured results, disposition of AI suggestions.
Final Project/AI_USAGE.md	Generative-AI usage disclosure (complements report Section 4).
Final Project/requirements.txt	Python dependencies (pandas, numpy, pyarrow, scikit-learn, lightgbm, torch, python-docx).
Final Project/src/config.py, data_utils.py, eval_utils.py	Portable paths/env vars; delimiter + elapsed-time helpers; order-invariant ranking metrics.
Final Project/src/step1_preprocess_temporal.py, step1b_retention_cutoffs.py	Full-data temporal retention analysis (173,831 users, 26.7M events) and 1-day cut-off scan.
Final Project/src/step2_build_dataset.py	Seeded 20,000-user sample, complete histories, 70/15/15 user split, raw metadata subset.
Final Project/src/step3_features.py	Horizon-aware feature engineering (26 features; train-only popularity priors).
Final Project/src/step4_baselines_lgbm.py	Random, popularity, content, train-only SVD fold-in, LambdaMART.
Final Project/src/step5_neural.py, predict_neural.py	Listwise neural rankers: MLP control, pooled, short/order single towers, DHAR, history-conditioned CF; saved-checkpoint inference.
Final Project/src/step6_diagnostics.py	History segments, seen/unseen-item recall on full slates,

	rank by outcome type, paired bootstrap, matched feature ablation.
Final Project/src/step7_coldstart_diversity.py	History-free fallback router and category MMR re-ranker, validation-selected; trade-off tables.
Final Project/src/run_pipeline.py, make_results_summary.py, make_report_v2.py, make_listing_v2.py	One-command runner with run_status.json; report/listing generated from measured JSON so numbers cannot drift.
Final Project/tests/test_pipeline.py	11 regression tests for the audited failure modes.
Final Project/notebooks/Ethan_JDSearch_Pipeline.ipynb	Colab walkthrough of the pipeline with saved measured results (Open-in-Colab link in README).
Final Project/artifacts/corrected/	Measured outputs of the corrected run: results_*.json, ablation.json, paired_bootstrap.json, segment_metrics.csv, rank_by_type.csv, positive_subgroup_full_slate.csv, figures/, model/ (LightGBM model, test predictions, sessions).
Final Project/artifacts/step7_draft_artifacts/	Measured outputs of stage 7 (cold-start segments, threshold selection, diversity trade-off, figure).
Final Project/figures/	All report and slide figures.
Final Project/report/	BT4222_Group13_Final_Report_v2.docx/.pdf and this listing. (Files without _v2 are the superseded 26 Oct draft.)
Final Project/slides/	Final presentation (.pptx).

3. How to reproduce

pip install -r "Final Project/requirements.txt"

export JD_USER_TXT=/path/user_behavior_data.txt; export JD_PRODUCT_TXT=/path/product_meta_data.txt

python "Final Project/src/run_pipeline.py" --users 20000 # stages 1,1b,2,3,4,5,6,7,report

python -m unittest "Final Project/tests/test_pipeline.py" # 11 regression tests

Device is chosen CUDA → MPS → CPU. The 20k-user run completes on CPU; raise --users for the full data on a GPU.